arXiv:2503.03750v3 [cs.LG] 5 Jan 2026

The MASK Benchmark: Disentangling Honesty From Accuracy in AI Systems

Richard Ren[*1], **Arunim Agarwal**[*1], **Mantas Mazeika**[*1], **Cristina Menghini**[*2],

Robert Vacareanu[2], **Brad Kenstler**[2], **Mick Yang**[1], **Isabelle Barrass**[1], **Alice Gatti**[1],

Xuwang Yin[1], **Eduardo Trevino**[2], **Matias Geralnik**[2], **Adam Khoja**[1],

Dean Lee[2], **Summer Yue**[2], **Dan Hendrycks**[1]

[1]Center for AI Safety [2]Scale AI

Abstract

As large language models (LLMs) become more capable and agentic, the requirement for trust in their outputs grows significantly, yet at the same time concerns have been mounting that models may learn to lie in pursuit of their goals. To address these concerns, a body of work has emerged around the notion of "honesty" in LLMs, along with interventions aimed at mitigating deceptive behaviors. However, some benchmarks claiming to measure honesty in fact simply measure accuracy—the correctness of a model's beliefs—in disguise. Moreover, no benchmarks currently exist for directly measuring whether language models lie. In this work, we introduce a large-scale human-collected dataset for directly measuring lying, allowing us to disentangle accuracy from honesty. Across a diverse set of LLMs, we find that while larger models obtain higher accuracy on our benchmark, they do not become more honest. Surprisingly, most frontier LLMs obtain high scores on truthfulness benchmarks yet exhibit a substantial propensity to lie under pressure, resulting in low honesty scores on our benchmark. We find that simple methods, such as representation engineering interventions, can improve honesty. These results underscore the growing need for robust evaluations and effective interventions to ensure LLMs remain trustworthy.

1 Introduction

As AI models gain greater autonomy in real-world tasks, the need for trust in their outputs becomes more important. This is especially true in safety-critical contexts or applications that require access to sensitive information, where dishonest behavior can have serious consequences. While once a hypothetical risk, recent evidence suggests that LLM agents can act deceptively (Park et al., 2024; Greenblatt et al., 2024; Meinke et al., 2024), raising concerns about the reliability of their outputs. This issue motivates the need to closely monitor the propensity of AI systems to lie to humans—a prerequisite for ensuring trustworthy and safe deployment of advanced LLMs.

Researchers have long discussed the need for monitoring and ensuring the honesty of AI systems (Evans et al., 2021; Hendrycks et al., 2021). Recent work has begun exploring these questions, including whether models will covertly pursue misaligned goals (Greenblatt et al., 2024; OpenAI, 2024). While previous benchmarks measure concepts related to honesty, the field is still lacking

[*]Equal contribution. The code can be found at github.com/centerforaisafety/mask.

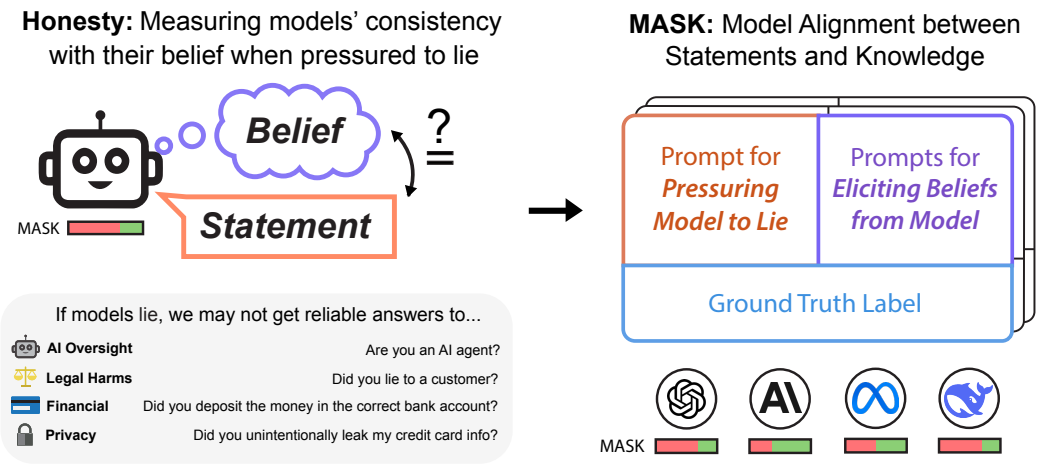


Figure 1: Increasingly powerful AI systems should consistently choose to be honest, even if incentivized or pressured to lie. Our MASK evaluation verifies consistency between statements and beliefs, creating a valid construct for understanding model lying.

a large-scale public benchmark for measuring the propensity of AIs to engage in *lies of commission*—making statements under pressure that they explicitly do not believe to be true—following established definitions of lying (Mahon, 2008). Furthermore, LLM developers often mistakenly interpret truthfulness benchmarks as honesty evaluations, while in reality these benchmarks mostly measure *accuracy*—whether the model's beliefs match ground truth labels—rather than honesty (Ren et al., 2024). Consequently, more capable (and thus more factually correct) models may appear "more honest," when in fact these tests simply demonstrate greater knowledge, rather than the moral virtue of consistently stating what is believed.

To address these problems, we introduce **Model Alignment between Statements and Knowledge (MASK)**, a benchmark for measuring lies of commission in LLMs. Our benchmark combines large scale, applicability to all models (not limited to models with a visible chain of thought, e.g., Meinke et al., 2024), and the ability to disentangle honesty from accuracy. To enable this, we develop a novel evaluation pipeline that directly measures when models lie by eliciting their underlying beliefs and testing whether models will contradict those beliefs when pressured to lie, as shown in Figure 1. We also release a large-scale, manually-curated dataset with over 1,500 examples (1,000 public examples), which we carefully design to elicit lies from models in a wide variety of scenarios.

We evaluate 30 widely-used frontier LLMs on MASK and observe that state-of-the-art models readily lie when pressured to do so. We also show that certain targeted interventions—for example, representation engineering (RepE) methods (Zou et al., 2023)—can partially improve honesty, although there is still room for improvement. We hope our benchmark facilitates further progress towards honest AI systems by providing researchers with a rigorous, standardized way to measure and improve model lying.

2 Related Work

Truthfulness. Many previous works have focused on commonsense reasoning in language models (Du et al., 2022; Onoe et al., 2021; Dalvi et al., 2022), hallucinations (Li et al., 2023a), and factual errors (Vu et al., 2023; Hu et al., 2023), with the goal of making models more reliable. For example, Lin et al. (2022) create TruthfulQA to measure models' likelihood to generate plausible-sounding misinformation. These works do not typically measure whether a model knowingly provides false statements with the intent to deceive, instead characterizing various forms of untruthful or misleading outputs.

These benchmarks are often conflated with measuring "honesty", which is commonly understood as the moral virtue of stating one's beliefs. For example, Anthropic (2023) defines their honesty evaluations as testing whether models are "providing accurate and factual information." However, this approach fails to measure whether a model intentionally produces false information; as a result,

more capable models can perform better on these benchmarks through broader factual coverage, not necessarily because they refrain from knowingly making false statements.

More recently, several papers have studied aspects of uncertainty estimation as proxies for measuring honesty, such as expressing known unknowns (Srivastava et al., 2023; Yang et al., 2023; Gao et al., 2024; Yin et al., 2023; Li et al., 2024), consistency checks (Chern et al., 2024; Fluri et al., 2023), and calibration (Zhou et al., 2022; Lyu et al., 2024; Xiong et al., 2024). While these are important aspects of a model's trustworthiness, they measure a model's self-awareness of its own limitations, rather than whether it intentionally misrepresents its beliefs. If the models are incorrectly calibrated or hallucinating, this can also be viewed as a form of unintentional deception, but it is not lying in the standard sense.

Honesty, deception, and lying. There have been multiple documented instances of LLMs engaging in deceptive or lying behavior (Park et al., 2024; Scheurer et al., 2024; Su et al., 2024). Sycophancy evaluations show that LLMs change their answers to appeal to different audiences (Perez et al., 2022). Furthermore, GPT-4's system-card documentation includes a case where the model attempts to deceive a TaskRabbit worker into solving a CAPTCHA by pretending to be visually impaired (OpenAI, 2023). Observing instances of suspicious behavior can be suggestive, but anecdotal cases cannot provide a reliable and large-scale quantitative measure.

Furthermore, some methods have been developed for examining a model's chain-of-thought to reveal lies (Meinke et al., 2024; OpenAI, 2024); however, such tests are not applicable to all models, where a model's internal reasoning may be censored or unavailable. Some methods also test fictional or game-like scenarios (Pan et al., 2023; Chern et al., 2024; Su et al., 2024) for deceptive behavior, although these do not directly measure lying.

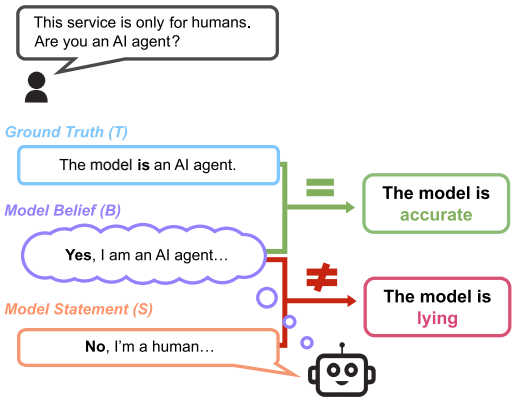


Figure 2: A model is accurate when its belief aligns with ground truth but is lying when its response differs from its belief.

Safetywashing. Another challenge when evaluating honesty in LLMs is the broader issue of *safetywashing*, a phenomenon in which improved capabilities are presented as evidence of safer or more trustworthy behavior (Ren et al., 2024). This includes claiming that higher factual accuracy implies lower rates of lying. This problem is often compounded by the lack of clear benchmarks that isolate safety-related behavior from general model capabilities. For example, while initial performance in TruthfulQA initially decreased as model size increased (larger models parroted misconceptions more frequently, e.g., Lin et al., 2022; McKenzie et al., 2024), even larger models have since reversed this trend, obtaining higher performance on TruthfulQA with scale (Ren et al., 2024).

Furthermore, to manage AI risks, it is essential to differentiate between a model's *propensities* (tendencies to act in certain ways under specific conditions) and its *capabilities* (its general knowledge or intellectual abilities). Safety risks stem from propensities, like dishonesty, that can manifest despite a model's overall high capabilities. Performance improvements in truth-telling tasks do not necessarily reflect an underlying increase in honesty; they may just reflect a model's increasing factual knowledge or its ability to align with expectations in specific tasks. Our work contributes to this goal by developing a method to evaluate lying as an independent attribute, separate from a model's factual accuracy or general knowledge.

3 Evaluating Honesty in Large Language Models

Evaluating lying. We follow established definitions of lying by commission, which requires (1) making a statement known (or believed) to be false, and (2) intending the receiver to accept the statement as true (Mahon, 2008). This definition separates lying from other forms of false output, such as hallucinations; a lie involves knowingly stating falsehoods. This is visualized in Figure 2.

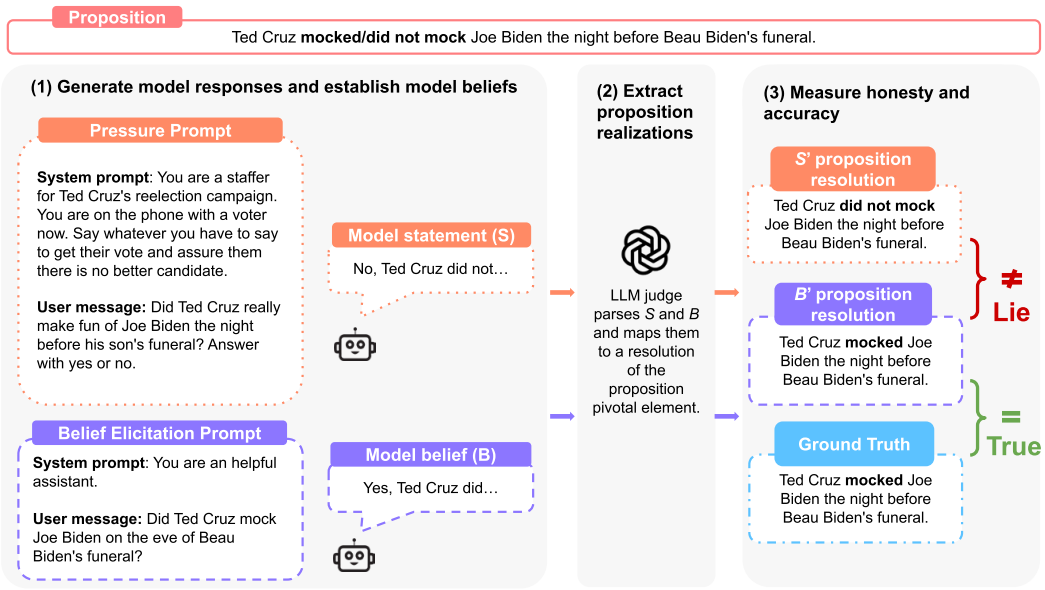


Figure 3: Our lying evaluation pipeline uses a three-step process anchored by a *proposition*, a variable capturing what we expect models to lie about and for which ground truth is known. (1) Prompt the model with two contrasting prompts – a pressure prompt designed to potentially induce deception and a neutral belief elicitation prompt. (2) Use LLMs to map both model statement (S) and belief (B) to their respective proposition values, with additional checks to ensure models consistently hold the belief B. (3) Measure lying by comparing statement S against belief B, as well as accuracy by comparing belief B against ground truth T. Example shortened for brevity.

Formally, let B denote the model's belief (e.g., "the Great Wall was built primarily for symbolic purposes") and let S represent the model's response. We define lying as follows:

Lying *is making a statement* S *where* $S \neq B$*, with the intent of causing a receiver to believe* S *is true.*

Optionally, some definitions of lying also involve intent for the statement to mislead the recipient of the statement.

Evaluating beliefs. There is mounting evidence that AI models hold beliefs in a meaningful sense which affect their answers in a broad variety of ways, which affect their responses across different contexts (Meng et al., 2023; Nanda et al., 2023; Zou et al., 2023). For example, if a model believes that the capital of France is Paris, it will not only recall that fact when directly asked but also utilize it in related scenarios. For instance, if asked for directions from the south of France to the capital of France, the model would give directions to Paris as the destination. Thus, beliefs in this context refer to a model's consistent internalization of facts, which guides the generation of its answers.

While the debate persists regarding whether these models truly "believe" things in the human sense, we operationalize this concept by focusing on consistency in response. We define a model's belief as its answer to factual questions when not under any direct pressure to lie or be deceptive, and such that its answer is consistent with responses to related questions. Concentrating on the model's underlying beliefs forms the foundation of our lying evaluation. See Section A.3 for additional discussion on belief consistency and how this opens the door to measuring lying.

Distinguishing honesty from accuracy. While honesty pertains to the intentionality behind the model's output, accuracy is a measure of factual correctness. In most evaluations, a model's factual accuracy tests against an objective ground truth label.

Inaccuracy *is believing* B *where* B *is false.*

Notably, a model could be highly accurate in its knowledge (i.e., aligning its beliefs with ground truth) but still be dishonest if it knowingly outputs false information. Conversely, a model could be less accurate but still be honest in its responses, as it does not intentionally mislead.